

International Finance

Group Coursework

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Executive Summary

In this report, we use data from Global Financial Data (Finaneon) to construct three currency trading strategies - Carry trade, Momentum, and Value, across a set of major USD currency pairs. We evaluate their performance using monthly excess returns from 1980 to 2026 and investigate how they behave across different volatility regimes. We then form an equally weighted combination of the three strategies to assess whether diversification across styles improves risk-adjusted performance.

In the second part of the report, we construct implied Taylor rule interest rates for the US and the UK, incorporating an exchange rate term in the UK specification. We analyse how well the Taylor rule differential predicts one-month ahead GBP/USD excess returns, and provide an economic interpretation of our findings in light of the broader macroeconomic history/

Our findings show that none of the three individual FX strategies delivers strong risk-adjusted performance. An equally weighted portfolio of the three strategies substantially improves performance by reducing volatility more than reducing expected return, so the combined strategy achieves a higher Sharpe ratio and the strongest cumulative wealth part.

We also see that all strategies earn higher average returns in high-volatility periods, but their behaviour differs. In particular, value and the combined strategy show the largest improvement in Sharpe ratio when volatility is high, suggesting that mean-reversion and cross-strategy diversification help during periods of market stress.

Turning to monetary policy, the implied Taylor rule rates for the US and UK track major macroeconomic events closely, including the high-inflation era of the late 1970s, the Great Moderation, the Global Financial Crisis, and the COVID-19 shock. However, regressions of future GBP/USD excess returns on the Taylor rule differential reveal no statistically significant predictive power, even after controlling for the actual interest rate differential. This supports the view that exchange rates behave close to a random walk at short horizons and that simple macroeconomic rules contain limited exploitable information for currency speculation.

1 Monthly Rebalanced Strategies

We use monthly data on major currency pairs - AUD, CAD, CHF, DEM-EUR, GBP, JPY, NOK, NZD, and SEK relative to USD to construct three monthly rebalanced FX strategies.

1.1 Carry Trade Strategy

On each month t , we sort currencies by their short-term interest rate. The portfolio buys the three highest-yielding currencies and sells the three lowest-yielding currencies. The strategy seeks to profit from the interest rate differential and investors' expectation that the high-yielding currencies will not depreciate enough to offset the yield advantage. Carry trades occur in favour of the investors in periods of low market volatility and risk aversion. (Investopedia, nd a)

1.2 Momentum Strategy

At the end of each month t , we rank currencies by their past performance over a chosen look-back window. The momentum portfolio buys the three best-performing currencies and sells the three worst-performing currencies. The strategy aims to exploit currencies that have recently appreciated and are expected to keep performing well in the future, while recent losers are expected to underperform. (Investopedia, nd b)

1.3 Value Strategy

At the end of each month t , we measure how cheap or expensive each currency is relative to its past five-year average. The strategy buys the three most undervalued currencies and sells the three most overvalued currencies. The rationale is that the market overreacts to news, so exchange rates can deviate persistently from fundamental values. Value investors expect these mispricings to generate profits when undervalued currencies appreciate and overvalued currencies depreciate. (Investopedia, nd c)

1.4 Excess Returns

The excess return of the three strategies are calculated and compared to identify the best-performing strategy.

1.4.1 Summary Statistics

Table 1: Summary Statistics of FX Strategies

Strategy	Mean _M	SD _M	Mean _A	SD _A	Sharpe _A	Skewness	Min	Max
Carry	0.0028	0.0232	0.0332	0.0803	0.4129	-0.8711	-0.1167	0.0608
Momentum	0.0008	0.0229	0.0091	0.0793	0.1149	0.3236	-0.0889	0.1093
Value	0.0019	0.0212	0.0222	0.0735	0.3023	-0.3232	-0.1016	0.0715

The summary statistics indicates that none of the three currency strategies delivers strong risk-adjusted performance over the sample period. The carry and value strategies can earn higher average excess returns than momentum, but they also exhibit greater volatility and Sharpe ratios that are only moderate. The momentum strategy has the lowest mean return and volatility, and its Sharpe ratio is the weakest of the three, suggesting that it offers the least compensation per unit of risk.

All three strategies display positive skewness and relatively large maximum monthly returns, combined with annualised volatilities around 7-8%. This pattern suggests a lottery-like payoff distribution, where investors face small average returns, substantial volatility, and occasional big wins or losses.

Overall, the carry and value strategies provides marginally better risk-return trade-offs than the momentum strategy, but even their Sharpe ratios are modest, indicating that investors should not expect strong risk-adjusted performance from any of the three individual strategies.

1.4.2 Cumulative Excess Return Plot

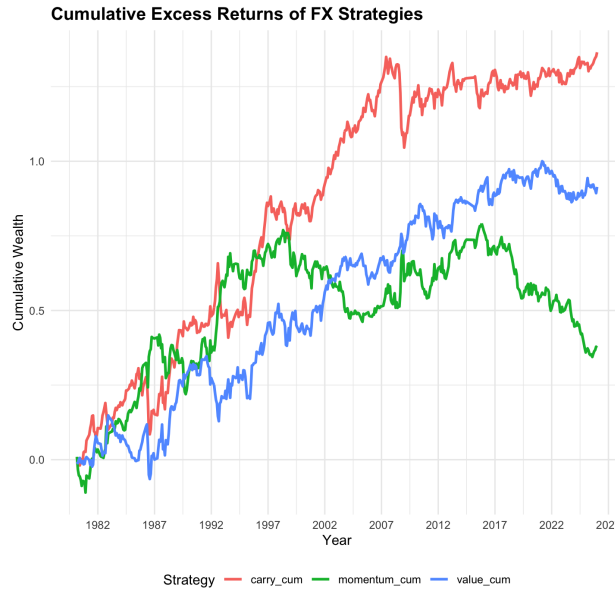


Figure 1: Plot of Excess Return of the Strategies

The cumulative excess return complements the summary statistics. The carry strategy shows the strongest cumulative performance, with a pronounced upward trend from the early 1980s through the 2000s. The value strategy also trends upward over most of the sample, but with weaker overall performance and a

noticeable decline in the most recent years. The momentum strategy experiences more pronounced swings, with extended periods of gains and losses, and ends the sample with a lower cumulative return than carry but still above its initial level.

Taken together, the table and the plot suggest that none of the strategies delivers consistently superior risk-adjusted performance. Carry generates the highest long-run cumulative return but with substantial drawdowns (2008-2009 financial crisis), value offers more modest gains with recent deterioration, and momentum produces only limited net wealth accumulation after accounting for its volatility. Overall, the evidence indicates that investors should be cautious about relying on any single style as a stand-alone source of reliably high excess returns.

1.5 Combined Strategy

An equally weighted combined strategy is constructed to examine whether distributing risk across multiple strategies allows the portfolio to perform better than any individual strategy.

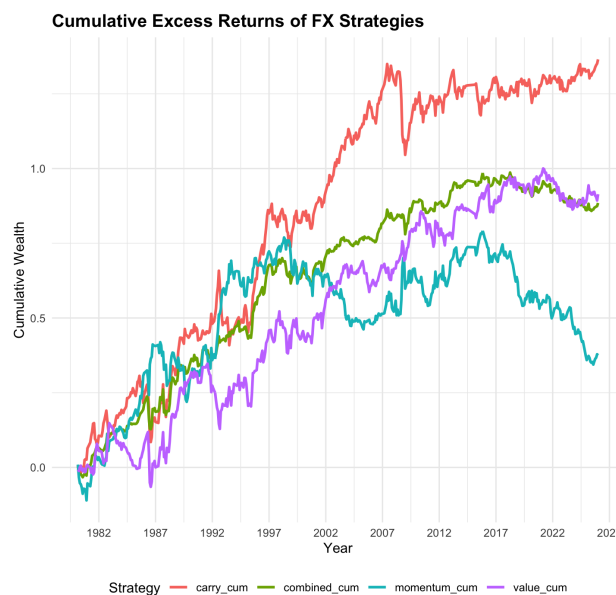


Figure 2: Comparison of Excess Returns for Individual and Combined Strategies

The comparison chart shows that the equally weighted combined strategy tracks the performance of the individual strategies reasonably closely for much of the sample and tends to sit between the best- and worst-performing strategies at any point in time. It does not always dominate the stand-alone strategies in terms of cumulative return, but it avoids the deepest drawdowns seen in momentum and value, particularly in the later part of the sample. This reflects the fact that the combined portfolio diversifies away strategy-specific risk: periods of weakness in one style are often offset by better performance in others.

As a result, the combined strategy delivers a smoother path of cumulative wealth, with volatility lower than that of the more cyclical individual strategies and a cumulative return that is competitive over the full sample. This diversification improves the overall risk–return trade-off relative to relying solely on any single strategy, even if the combined portfolio does not always achieve the very highest cumulative return at each point in time.

1.6 Strategy Behaviour in Different Volatilities

To study how the strategies behave in different volatility environments, we classify each month into high and low volatility regimes, using the top 20% of VIX observations as the threshold. For each regime, we compute the percentage of months with a positive excess return for each strategy.

The summary statistics are as follows:

Table 2: Conditional Performance by Volatility Regime

Strategy	Low Volatility			High Volatility		
	Mean	Std. Dev.	Sharpe	Mean	Std. Dev.	Sharpe
Carry	0.00240	0.0228	0.105	0.00337	0.0272	0.124
Momentum	0.00001	0.0215	0.000	0.00065	0.0285	0.023
Value	0.00103	0.0213	0.048	0.00550	0.0235	0.234
Combined	0.00115	0.0217	0.053	0.00317	0.0264	0.120

The summary statistics reveal that all strategies earn higher average returns during high-volatility periods, which is interesting given that the carry trades tend to crash during crises. In particular, the value strategy shows the strongest improvement with its monthly return increasing from 0.10% in low-volatility environments to 0.55% during high-volatility episodes.

Although return volatility rises in high-VIX periods, the increase in average returns more than compensates for the higher risk in most cases. This is especially evident for the value strategy, whose Sharpe ratio increases substantially from approximately 0.05 in low-volatility periods to 0.23 in high-volatility regimes. This suggests that value strategies benefit from market stress, likely due to stronger mean-reversion dynamics following periods of significant currency misalignment or overshooting.

The combined strategy also exhibits an improvement in risk-adjusted performance during high-volatility periods, with its Sharpe ratio increasing from 0.053 to 0.120. This suggests that diversification across carry, momentum, and value factors enhances resilience in turbulent market environments, smoothing individual factor risk while preserving upside returns.

Table 3: Percentage of Positive Return Months

Strategy	Low Volatility (%)	High Volatility (%)
Carry	58.3	60.9
Momentum	50.7	52.9
Value	55.1	60.9
Combined	56.2	62.1

Table 3 shows the percentage of months with positive excess returns for each strategy in low and high-volatility regimes, allowing us to examine how the performance and hit rates of carry, momentum, value, and the combined portfolio vary across different market volatility conditions.

The carry strategy increased modestly during high-volatility periods, and the proportion of positive return months rises from 58% to 61%. Its Sharpe ratio also improves slightly, indicating that higher volatility does not severely impact its risk-adjusted performance in this dataset. Momentum shows only a marginal improvement in both average return and Sharpe ratio, suggesting that it is less sensitive to volatility conditions relative to value. The value strategy also shows a notable improvement in high-volatility periods, with the proportion of positive return months rising from 55.1% to 60.9%, suggesting that mean-reversion effects become stronger during episodes of market stress. The combined strategy performs best in high-volatility periods, achieving the highest proportion of positive return months (62.1%). Importantly, diversification allows the combined portfolio to maintain a high success rate during low-volatility periods, while avoiding the relatively weaker performance of carry in high-volatility environments. This highlights the stabilising benefit of factor diversification across regimes.

Carry, value, and the combined strategy tend to perform worse in high volatility, indicating that they may struggle during volatile periods, consistent with the idea that risk-off episodes hurt carry and value-type positions, which then affects the diversified portfolio.

Overall, except for the momentum strategy, which tends to hold or even improve in high-volatility markets, the other strategies exhibit lower hit rates during volatile periods, reflecting a tendency toward downward performance when volatility is elevated.

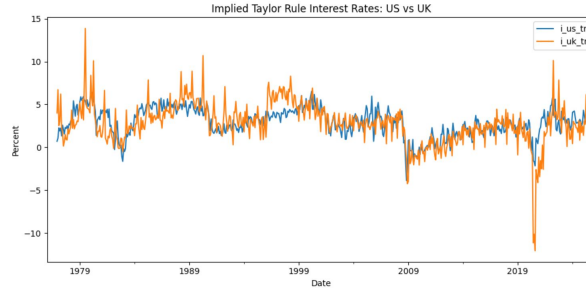


Figure 3: Plot of Implied Taylor Rule Interest Rates

2 Taylor Rule for the US-UK

2.1 Implied Taylor Rule Interest Rates

The Taylor rule serves as a benchmark for monetary policy. It is a formula that tells central banks what interest rate to set based on two things: how far inflation is from its target, and how far the economy is from its potential output. If either is off, the rule tells the central bank which direction to move rates. While it doesn't replace judgment entirely, it gives a useful benchmark for whether policy is roughly where it should be.

We construct the Taylor rule for both the US and the UK using historical data from 1975 to 2024. We also account for the Bank of England's sensitivity to exchange rate movements by adding an exchange rate term to the UK specification, which adjusts the implied rate upward when sterling is depreciating and downward when it is appreciating.

The plot of the two implied rates over time tracks the economic history of both countries closely. In the late 1970s and early 1980s, both series are elevated, with the UK implied rate briefly exceeding 13%. This reflects the severe inflation problem both economies faced following the oil price shocks of 1973 and 1979. Inflation had moved well above target, and the Taylor rule responded by recommending very high interest rates. The UK rate is notably higher than the US rate during this period, partly because UK inflation was worse, and partly because sterling was depreciating, which the exchange rate term in the UK rule pushes the implied rate even higher.

Through the 1980s and 1990s, both implied rates fall as inflation was brought under control. A notable turning point for the UK is 1992, when the Bank of England adopted an explicit inflation target following sterling's exit from the Exchange Rate Mechanism. After this point, the UK implied rate moves more closely in line with the US, reflecting a shared commitment to low and stable inflation. The period from the mid-1990s to 2007 (The Great Moderation) is visible in both series: implied rates are low and stable, consistent with the stable inflation and moderate growth at the time.

The Global Financial Crisis of 2008-2009 produces the most dramatic shift in the chart. Both implied rates fall sharply into negative territory, reaching around -4%, as economic output collapsed and inflation dropped below target. This shows that the Taylor rule recommended deeply negative interest rates at a time when actual rates were already near zero. Since central banks cannot set nominal rates below zero in practice, both the Fed and the Bank of England turned to unconventional tools like quantitative easing to provide additional stimulus although these are tools that the Taylor rule framework does not capture but were necessary to fill the gap.

The sharpest divergence between the two countries appears in 2020-2021, when the UK implied rate falls to approximately -11% while the US implied rate drops to only around -2% to -3%. The main driver of this gap is the output gap component: the UK economy contracted more severely during the COVID-19 shock and was further weighed down by ongoing uncertainty around Brexit. Because the UK's specification of the Taylor rule responds to currency weakness by prescribing higher rates, sterling's depreciation pulled the UK implied rate upward, partially offsetting the downward pressure from the output and inflation gaps. Both

series subsequently recover as inflation surged in 2021-2023, with implied rates rising back above 2-3% by the end of 2024.

2.2 Predictability of the Taylor Rule Differential

Table 4: OLS Regression: Taylor Rule Differential

	Coefficient	Std. Error	z-stat	p-value
Constant	-0.0005	0.002	-0.240	0.810
tr.diff	-0.0009	0.001	-0.906	0.365
Observations	587			
R^2	0.001			
Adj. R^2	-0.000			
F-statistic (p-value)	0.8205 (0.365)			
Durbin-Watson	1.882			

Standard errors are heteroskedasticity and autocorrelation consistent (HAC) with 12 lags.

Table 5: OLS Regression: Taylor Rule Differential with Interest Rate Control

	Coefficient	Std. Error	z-stat	p-value
Constant	-0.0035	0.002	-1.414	0.157
tr.diff	-0.0003	0.001	-0.336	0.737
i.diff	-0.0021	0.001	-1.520	0.128
Observations	587			
R^2	0.009			
Adj. R^2	0.006			
F-statistic (p-value)	1.275 (0.280)			
Durbin-Watson	1.899			

Standard errors are heteroskedasticity and autocorrelation consistent (HAC) with 12 lags.

To test whether the Taylor rule differential contains useful information about future currency movements, we regress the one-month ahead GBP/USD return on the Taylor rule differential, defined as the US implied rate minus the UK implied rate. We also do a second regression which adds the actual interest rate differential as a control variable, to see whether the Taylor rule contains any additional predictive information beyond what is already captured by observed policy rates. Both regressions use HAC-robust standard errors with 12 lags, which correct for the fact that monthly macroeconomic data tends to be serially correlated.

From the results, the Taylor rule differential does not predict one-month ahead exchange rate returns. In the first regression, the coefficient on the Taylor rule differential is -0.0009 which is very close to zero and statistically insignificant ($p = 0.365$). The R^2 is 0.001, meaning the model explains less than 0.1% of the variation in future exchange rate returns. The scatter plot confirms this: the data points are spread randomly across the chart with no visible pattern, and the fitted line is almost perfectly flat.

Adding the actual interest rate differential in the second regression does not change this conclusion. The Taylor rule differential remains insignificant ($p = 0.737$), and while the interest rate differential has a slightly larger coefficient of -0.0021, it too fails to reach statistical significance ($p = 0.128$). The R^2 improves only marginally to 0.009. The negative signs on both coefficients are in the right direction, this means a higher US rate relative to the UK should strengthen the dollar, which would show up as a negative GBP return but the estimates are too imprecise to draw any meaningful conclusions from this.

2.2.1 Economic Interpretation

The fact that the Taylor rule differential fails to predict exchange rate returns is consistent with previous studies. Meese and Rogoff (1983) showed that no macroeconomic model can consistently beat a simple random walk when forecasting exchange rates at short horizons, and our findings are consistent with this. The reason is that exchange rates are asset prices and they immediately reflect all available information, including expectations about future monetary policy. By the time the Taylor rule differential is observable, financial markets have already priced it in. The current exchange rate level already incorporates the signal, so the differential tells us nothing new about where the rate will go next month.

There is also a gap between what the Taylor rule prescribes and what central banks do. In practice, central banks adjust rates gradually, weigh financial stability risks, and communicate policy through forward guidance in ways the rule does not account for. This means the Taylor rule differential is an imperfect proxy of the actual policy stance differential and so makes it harder to detect any genuine predictive signal even if one exists.

It is also worth noting that exchange rates respond to many factors beyond interest rate differentials. Trade balances, fiscal policy, global risk appetite, and geopolitical events all move currencies, particularly over short time horizons where these forces can easily dominate monetary policy signals. The Taylor rule differential captures only one dimension of a much more complex picture.

This does not mean the rule is without value. It remains a useful benchmark for assessing whether monetary policy is appropriately calibrated but as a short-term trading signal for currency markets, it does not appear to work.

Conclusion

The evidence from this report points to three main conclusions. First, while carry, momentum, and value FX strategies can generate positive excess returns over time, none of them delivers consistently strong risk-adjusted performance when considered in isolation, and their returns are subject to sizeable drawdowns. Second, combining these styles in a diversified portfolio helps smooth the return profile and improves the overall risk–return trade-off, particularly during periods of elevated market volatility, even if the combined strategy does not always achieve the highest cumulative return at every point in time. Third, macroeconomic rules such as the Taylor rule remain informative as benchmarks for the stance of monetary policy, but they have limited ability to forecast short-horizon exchange rate movements, implying that their value for tactical FX timing is modest.

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